VisionEV: Multimodal Large Language Models for Spatially Aware Electric Vehicle Charging Demand Prediction Using Satellite Imagery

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Abstract: This manuscript presents VisionEV, a multimodal large language model (LLM) framework designed to predict electric vehicle (EV) charging demand by leveraging satellite imagery and structured textual data. Spatial context—including parking availability, land use, and nearby amenities—is critical for accurate demand estimation. However, site selection in EV infrastructure planning remains both labor-intensive and inconsistent, requiring human experts to conduct in-person audits and manually define spatial features. To overcome these limitations, VisionEV introduces an automated spatial reasoning pipeline that integrates satellite imagery of candidate locations as visual inputs, allowing the model to learn nuanced spatial patterns directly from imagery, without relying on predefined descriptors. Complementary station-level attributes, including traffic flow and temporal indicators, are embedded into domain-informed textual prompts to simulate planner reasoning. A core technical challenge lies in enabling coherent reasoning across semantically distinct inputs—structured textual data and perceptual visual context. VisionEV addresses this by reformulating the task as multimodal text generation, aligning both modalities within a shared embedding space through vision-informed prompting and lightweight domain-adaptive fine-tuning. We evaluate VisionEV using a real-world dataset of 22,852 training samples and 2,858 test samples collected from 189 public EV charging stations in Kansas City, Missouri. In the full-shot setting, VisionEV achieves superior accuracy (RMSE: 2.87, MAE: 1.98), outperforming the strongest baseline, LightGBM, by 1.0% and 5.3%, respectively. Few-shot, within-city zero-shot and cross-region spatial hold-out experiments demonstrate VisionEV's ability to generalize across unseen scenarios, and ablation studies confirm the contributions of visual input, prompt design, and fine-tuning. These results underscore the promise of multimodal LLMs in supporting scalable, data-driven EV infrastructure planning through automated spatial understanding.

Keywords: Electric vehicle; Charging demand prediction; Multimodal large language model; Multimodal learning; Satellite imagery

1. Introduction

The global transition toward electric vehicles (EVs) is accelerating as governments and industries prioritize decarbonization and sustainable mobility. While this shift brings substantial environmental benefits, it also creates significant challenges for infrastructure planning, particularly in the strategic deployment of public charging stations. Accurate forecasting of EV charging demand is vital to align infrastructure investments with actual usage patterns, ensuring that stations are built where and when they are needed. From an operational standpoint, demand prediction enables charging providers to allocate resources efficiently, minimize queuing, and alleviate driver range anxiety. From an energy systems perspective, it supports grid operators in managing spatiotemporally variable loads introduced by EVs, helping prevent local distribution bottlenecks (Wang et al. 2023). Without reliable forecasts, planners risk under-provisioning in high-demand areas or overbuilding in low-demand zones, leading to wasted capital and poor service outcomes. As such, charging demand prediction plays a foundational role in enabling scalable, cost-effective, and user-centric EV infrastructure development (Song and Hu 2023).

Numerous prior studies have demonstrated the critical role of spatial context in accurately predicting EV charging demand. Charging behavior is shaped not only by temporal patterns but also by environmental factors such as parking availability, proximity to commercial hubs, residential density, roadway accessibility, land use types, and the presence of nearby amenities. Effectively capturing this spatial heterogeneity is essential for robust demand forecasting, especially in complex urban environments. To account for these influences, many recent studies incorporate manually defined geographic features into their forecasting models. For example, Shi et al. (2023) applied spatiotemporal graph convolutional networks to model inter-station dependencies and temporal dynamics, while Kuang et al. (2024) proposed a physics-informed graph learning framework that incorporates road density, POI distribution, and pricing data. These efforts demonstrate that effectively encoded spatial features can substantially enhance demand forecasting accuracy. However, they typically rely on human-defined variables, which may not fully capture the richness and complexity of the built environment.

In-person site visits are required in practice to comprehensively identify and document relevant spatial features that influence EV charging demand. Such visits constitute an essential step in EV infrastructure planning by enabling planners to capture detailed, site-specific information not readily available through standard geospatial databases. For example, in a study in Kansas City, Missouri (Pan et al. 2025), planners systematically observe and record conditions including the availability and accessibility of parking spaces near potential charging locations, characteristics of surrounding land use (e.g., residential, commercial, industrial), and the overall visibility and convenience of candidate sites for users. Additionally, planners carefully evaluate contextual constraints such as the proximity of proposed charging stations to residential frontages, critical urban infrastructure like fire hydrants, and intersections to ensure safety and compliance with local regulations. These meticulously conducted observations help ensure that chosen locations effectively meet anticipated usage patterns and are compatible with surrounding urban contexts, thereby supporting informed and sustainable infrastructure development decisions.

In Figure 1 below, *Approach 1* illustrates the traditional EV charging site selection process. Planners typically begin by manually defining features such as network geometry and annual average daily traffic (AADT), which are then fed into a demand prediction model—based on either simulation or machine learning—to identify and rank candidate sites. High-ranking sites are subsequently audited through in-person visits, during which detailed spatial attributes are observed, quantified, and incorporated into a refined demand prediction model to support final site selection decisions. However, this approach, centered on human-led site visits and inspections, faces two significant limitations:

1. Labor-Intensive and Resource-Constrained: Site visits require significant time, personnel, and financial investment, limiting the number of candidate locations planners can feasibly evaluate. For instance, a prior study documents that planners initially identified approximately 300 potential charging locations

but had to narrow the selection to just 60 sites for detailed in-person evaluation (Pan et al. 2025), inevitably overlooking potentially valuable candidates.

2. Subjectivity and Arbitrary Feature Selection: The accuracy and completeness of spatial data collected during site visits depend heavily on planners' professional experience and subjective judgment. Planners must decide which specific spatial variables to identify and quantify, introducing inherent arbitrariness. Without comprehensive expertise and meticulous due diligence, crucial spatial features influencing charging demand could easily be overlooked, and even identified features may be inconsistently quantified, weakening the reliability and comparability of planning data.

Recent advances in large language models (LLMs) offer a promising alternative to traditional, manually engineered approaches. By reasoning over heterogeneous data in a unified semantic space, LLMs are capable of capturing complex dependencies without the need requiring explicit feature construction. For instance, ChatEV reformulates the EV charging prediction task as a natural language generation problem, embedding structured inputs, such as road density and pricing, into a prompt-based text representations (Qu et al. 2024). Such LLM-based approach has demonstrated strong predictive performance, particularly in data-limited scenarios. By abstracting the modeling pipeline into prompt engineering and language generation, LLMs reduce the need to manually define variable relationships, select model architectures, and carry out intensive training or tuning procedures. This streamlining significantly simplifies the demand forecasting workflow for candidate sites. The general process employed by ChatEV and similar LLM-based frameworks is summarized in Figure 1 - *Approach 2,* highlighting its distinction from both traditional pipelines and our proposed method. However, this LLM-assisted approach still relies on in-person site visits to collect site-specific spatial context, as language-based models are well-suited for processing textual inputs but struggle to capture spatial information that is typically conveyed through images rather than text.

In this study, we introduce VisionEV, a multimodal LLM-based forecasting framework that integrates structured textual inputs with spatial context derived from satellite imagery. Unlike existing LLM-based methods such as ChatEV, VisionEV directly perceives and extracts spatial features through publicly accessible satellite images, such as those available via the Google Maps API. This capability eliminates the need for manual, resource-intensive site visits, significantly simplifying and streamlining the EV site selection process, as illustrated in Figure 1 - Approach 3. The use of satellite imagery marks a significant paradigm shift: planners can now leverage both structured station-level data and automatically extracted spatial context from candidate sites. This allows for the comprehensive quantification and evaluation of a larger set of potential locations, eliminating the need for early-stage down-selection and iterative modeling, and ultimately improving the scalability and efficiency of EV infrastructure planning.

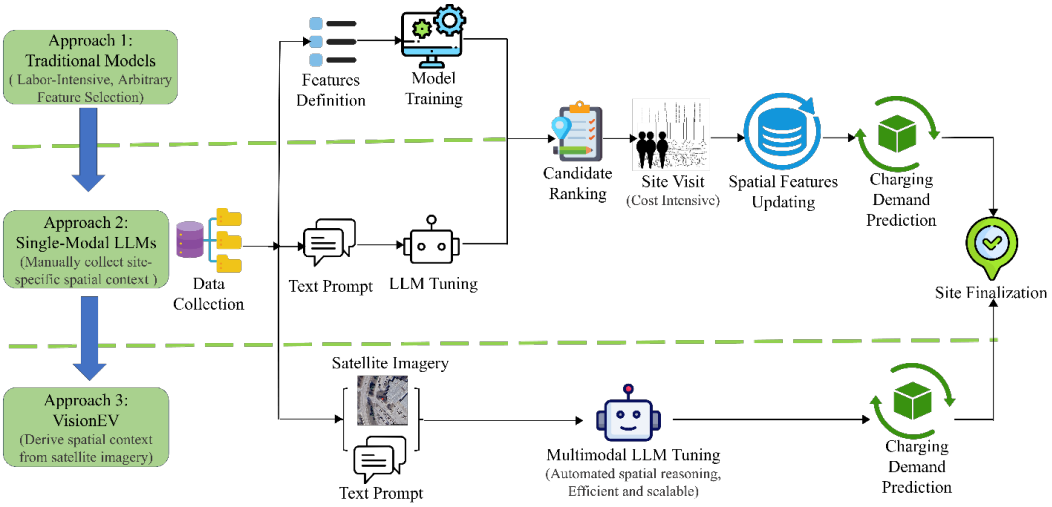


Figure 1 Comparison of Charging Demand Prediction and Site Selection Approaches

Exploring multimodal LLMs for forecasting EV charging demand poses significant challenges that have yet to be fully addressed. Unlike typical vision-language tasks such as image captioning or visual question answering, EV demand forecasting requires integrating symbolic station-level attributes, such as traffic volume and charger properties, with rich visual signals extracted from satellite imagery. A key challenge lies in aligning these semantically distinct modalities: structured textual inputs convey abstract numerical and categorical information, while unstructured images encode spatial context in a dense, perceptual format. Enabling coherent reasoning across these inputs demands a joint embedding space and task formulation that preserves cross-modal relationships. Compounding this, domain adaptation poses another layer of complexity. Multimodal LLMs like PaliGemma are pretrained on general-purpose internet data and lack inductive biases specific to infrastructure planning or mobility forecasting. These technical barriers underscore the need for tailored multimodal architectures that unify perception and reasoning in a domain-aware fashion—precisely the focus of the VisionEV framework introduced next.

To this end, VisionEV introduces two core modeling innovations to address these challenges: (1) automated visual spatial feature extraction and (2) structured prompt engineering. Specifically, we reformulate the EV charging demand prediction task as a multimodal text-generation problem, where publicly available satellite imagery around candidate stations is utilized directly as unstructured visual input. These images are automatically processed to capture critical spatial patterns such as land-use type, building density, and accessibility, thus eliminating the dependency on predefined spatial metrics. Concurrently, structured station-level attributes—including traffic flow data and temporal indicators—are embedded into carefully designed natural-language prompts that simulate the domain-specific reasoning processes of EV infrastructure planners. These multimodal inputs are then integrated into a shared embedding space using a vision-language backbone based on the PaliGemma model (Beyer et al. 2024), which supports joint reasoning over perceptual and semantic features. To further enhance model performance and domain adaptation, we implement a short-run, lightweight fine-tuning strategy using a cosine learning rate schedule. This approach ensures efficient integration of domain-specific knowledge into the forecasting pipeline, allowing effective and scalable prediction of EV charging demand without human-intensive spatial audits.

To rigorously assess VisionEV, we conduct extensive experiment on a real-world dataset from Kansas City, Missouri, comprising 22,852 station-week training samples and 2,858 held-out test samples. In the full-shot setting, VisionEV attains a Root Mean Squared Error (RMSE) of 2.87 and Mean Absolute Error (MAE) of 1.98, surpassing the strongest baseline, LightGBM (RMSE: 2.90, MAE: 2.09), by 1.0% and 5.3%, respectively. In addition, we evaluate VisionEV under few-shot, zero-shot settings and cross-region spatial hold-out to examine its robustness in limited-data scenarios. The few-shot setting highlights the model's ability to learn effectively from sparse training data, while the zero-shot setting and cross-region spatial hold-out validation demonstrates its generalization capability across unseen spatial contexts. Complementary ablation studies confirm that satellite imagery, prompt engineering, and lightweight fine-tuning each contribute materially to this performance gain.

The key contributions of this work are:

- Automated perceptual spatial representation: VisionEV is the first EV-demand predictor that replaces labor-intensive site visits with readily available satellite imagery, enabling the model to learn land-use mix, building density, and accessibility cues without handcrafted geographic features.
- Unified multimodal reasoning architecture: We reformulate charging-demand forecasting as a multimodal text-generation task and fuse visual embeddings with structured station attributes through domain-specific prompt engineering atop a vision-language backbone, allowing joint perceptual–semantic reasoning in a single forward pass.
- Lightweight, domain-adaptive fine-tuning strategy: A short-run cosine-scheduled fine-tuning procedure efficiently injects EV-specific knowledge into the backbone while preserving general vision-language capabilities, delivering strong accuracy with modest computational cost.
- Comprehensive empirical validation: Full-shot benchmarking against classical machine-learning, graph-based, and text-only LLM baselines—together with few-shot, zero-shot, cross-region spatial

hold-out and controlled ablations—demonstrates VisionEV's superior accuracy and scalability for large-scale site screening.

- Practical implications for infrastructure planning: By automating spatial feature extraction and eliminating manual audits, VisionEV enables planners to evaluate far more candidate sites, accelerating equitable and cost-effective EV-charging deployment in data-sparse regions.

The remainder of this paper is organized as follows. Section 2 reviews related work on EV charging demand forecasting and LLM-based prediction methods. Section 3 describes the proposed modeling framework, including the multimodal reasoning architecture, automated visual spatial feature extraction, structured prompt engineering and a short-run, lightweight fine-tuning strategy. Section 4 presents our case study in Kansas City and evaluates the model against baseline methods as well as variant models for ablation study. Section 5 concludes with a summary of findings.

2. Literature Review

Forecasting EV charging demand is essential for infrastructure planning, grid resilience, and user-centric service deployment. However, demand patterns are inherently complex—shaped by heterogeneous driver behaviors, mobility needs, and location-specific conditions such as land use or station accessibility. As EV adoption grows, these spatiotemporal dynamics introduce significant uncertainty into infrastructure planning. A wide array of predictive models has emerged in response, evolving from traditional statistical methods to advanced deep learning and, more recently, large language and multimodal frameworks.

Early research on EV charging demand prediction relied on statistical approaches such as autoregressive integrated moving average (ARIMA) and its seasonal variants (SARIMA), which model linear temporal dependencies. For example, Amini et al. (2016) applied ARIMA to forecast charging load under growing EV adoption and observed declining performance as variability increased. Regression-based models—both linear and polynomial—were also explored, often incorporating temporal or environmental covariates like time of day or temperature (Majidpour et al. 2014, De Cauwer et al. 2015). While these approaches offered interpretability, they were limited in capturing non-linear relationships, sparse station-level usage, and the sharp peaks commonly observed in real-world EV charging data. Furthermore, analytical models and queueing theory-based forecasts often lacked validation with empirical data, restricting their practical utility. Overall, statistical methods provided foundational insights but fell short in addressing the high variability and spatial complexity of EV charging behavior.

To address nonlinearity and improve forecasting performance, researchers turned to classical machine learning models. Techniques such as support vector regression (SVR), random forests, and gradient boosting trees (e.g., XGBoost) offered better handling of irregular patterns. For instance, Buzna et al. (2019) showed that tree-based models outperform basic time-series baselines when equipped with well-engineered features. However, these models still require manual feature selection for variables like time, weather, and location—which is labor-intensive and demands expert knowledge.

Deep learning methods, particularly recurrent neural networks (RNNs) and long short-term memory (LSTM) models, emerged as the dominant paradigm for capturing temporal dependencies in EV usage data. LSTM-based models demonstrated superior performance over traditional methods for short-term charging load prediction (Zhu et al. 2019), especially in capturing seasonality and long-range patterns. Variants such as gated recurrent units (GRUs) and hybrid CNN-LSTM architectures further improved accuracy by combining local feature extraction with sequential learning (Gunasekaran et al. 2024, Li et al. 2024). Despite their strength in modeling time-series data, most of these models still treated each station as an isolated unit or required predefined spatial clusters, which limited their ability to capture nuanced geographic relationships.

Recent advances in graph neural networks (GNNs) enabled more explicit modeling of spatial dependencies among charging stations. In these methods, stations are represented as graph nodes, with edges denoting physical or functional proximity (You et al. 2024). Spatiotemporal GNNs such as graph convolutional

networks (GCNs) combined with temporal layers have been shown to outperform LSTM models by accounting for shared patterns among nearby stations (Li et al. 2024). Extensions like adaptive graph learning (e.g., constructing adjacency matrices from latent correlations) aimed to overcome the rigidity of fixed graph structures. While GCN-based models excel in encoding networked relationships, they still depend on human-defined inputs such as road networks, distance thresholds, or functional adjacency—limiting their ability to learn high-level spatial semantics like neighborhood typologies or land use variations.

The most recent frontier reimagines EV demand forecasting as a natural language or multimodal learning task. Large language models offer a unique advantage: they can reason over heterogeneous data types when framed in a shared textual or semantic space. Qu et al. (2024) introduced ChatEV, a pioneering model that reformulates EV demand prediction as a text-to-text generation task. By embedding structured features—such as pricing, weather, and occupancy—into textual prompts, ChatEV leverages the semantic flexibility of transformer models to improve generalization. Hybrid architectures further extend the capabilities of LLMs. For example, integrating GNNs with LLMs has been shown to enhance spatial reasoning in EV optimization tasks (Orfanoudakis et al. 2025). In this setup, GNNs encode relational station data, while LLMs manage sequential decision-making—highlighting the potential of cross-domain integration. Still, these models rely on curated textual or graph-based representations of space, and none fully address the challenge of autonomously extracting spatial context from raw environmental inputs. In addition, a recent survey has provided a comprehensive overview of the integration pathways, applications, and challenges of large language models and AI-generated content technologies across complex system domains, offering valuable insights for extending such models to infrastructure and mobility applications(Guo et al. 2024).

Emerging multimodal approaches address this gap by integrating structured station attributes with unstructured inputs such as satellite imagery. Although research in this area is still nascent, related domains suggest its promise. For instance, remote sensing and urban planning studies have used satellite images to infer land use, parking availability, and built environment features. By leveraging multimodal LLMs like PaliGemma, it becomes feasible to incorporate this visual information into EV demand models—allowing the prediction task to be informed by perceptual representations of spatial context without requiring handcrafted spatial features.

To summarize, the literature shows a clear evolution from linear statistical models to machine learning, deep learning, and graph-based frameworks. Each stage brought greater modeling power and capacity to capture the complex dynamics of EV charging behavior. However, most prior methods still depend on human-defined and manually-extracted representations of space—whether through features, graphs, or clusters—which is labor-intensive and demands expert knowledge. Recent efforts using LLMs and prompt engineering have begun to address heterogeneity in a unified semantic space, but spatial context is still encoded manually.

This study addresses that gap by introducing VisionEV, a multimodal LLM-based framework that combines satellite imagery and structured data. Our approach eliminates the need for handcrafted spatial features by allowing the model to learn directly from raw visual data. Structured prompts guide the LLM to interpret station-level attributes while satellite images provide a perceptual layer of spatial context. Together, these modalities enable more accurate EV demand forecasting, particularly for diverse urban environments. This represents a natural progression in the literature and offers a new direction for spatially aware forecasting systems.

3. Methodology

For ease of reference, the mathematical notations used in this manuscript are listed below.

Category	Symbol	Description
	D	Training dataset with multimodal observations.
Data & Indices	N	Total number of station-week observations.
	i , w	Index: station, week.
	$I_i \in \mathbb{R}^{H\times W\times C}$	Satellite image for station i, as $\mathbb{R}^{H\times W\times C}$, H, W, C are height, width, channels.
Inputs	$\mathbf{s}_{i,w} \in \mathbb{R}^{d}$	Structured planning vector for station i at week w; d: # of features.
	$d_{i,w}$, $\hat{d}_{i,w}$	Observed and predicted weekly charging demand.
	$\phi_v(\cdot)$	Vision encoder (e.g., SigLIP)
	$\mathbf{v}_i \in \mathbb{R}^{n_v\times \mathrm{d}_v}$	Visual patch embeddings; n_v: patch count, d_v: embedding dim.
Image Feature	$\mathbf{W}_v \in \mathbb{R}^{d\prime\times \mathrm{d}_v}$	Projection matrix to token embedding space.
	$\mathbf{z}_i^{(img)} \in \mathbb{R}^{n_v\times d\prime}$	Projected visual embeddings.
	$\mathbf{T}_{i,w}$	Text prompt derived from $\mathbf{s}_{i,w}$.
Text Features	$\phi_t(\cdot)$	Text tokenizer/encoder.
	$\mathbf{z}_{i,w}^{(txt)} \in \mathbb{R}^{n_t\times d\prime}$	Text token embeddings; n_t: token count.
	$\mathbf{Z}_{i,w} \in \mathbb{R}^{(n_v+n_t)\times d\prime}$	Concatenated multimodal embeddings
	$Tok(\cdot)$,$DeTok(\cdot)$	Tokenization/detokenization functions
Fusion & Tokens	V	Token vocabulary
	$y_{i,w}$, $\hat{y}_{i,w}$	Tokenized observed and predicted demand.
	$f_\theta(\cdot)$	Multimodal model parameterized by θ.
	$\mathcal{L}(\theta)$	Negative log-likelihood loss.
Model & Training	B, E, T	Batch size, epochs, total steps
	η, η_t	Initial and dynamic learning rate.
	α	Warm-up ratio.

Category	Symbol	Description
	$\nabla_\theta \mathcal{L}(\theta)$	Loss gradient
	D^{train}, D^{val}	Training/validation sets

3.1 Overview

This section introduces VisionEV, a multimodal large language model framework designed for spatially-aware prediction of EV charging demand. VisionEV combines structured planning features and satellite imagery into a unified multimodal representation, enabling transformer-based models to perform accurate numerical forecasting. The methodological approach involves transforming diverse spatial and contextual inputs into a language-based sequence that incorporates both explicit and implicit spatial cues.

As illustrated in Figure 2, VisionEV operates through a structured five-stage pipeline: data inputting, prompting, embedding, fine-tuning, and demand prediction. Initially, the model collects multimodal data for public EV charging stations, including (a) historical charging log data with station occupancy and location coordinates; (b) satellite images from Google Maps API, capturing visual context like land use patterns, road networks, and urban form; and (c) travel demand model outputs such as AADT and trip production, indicating localized mobility and accessibility.

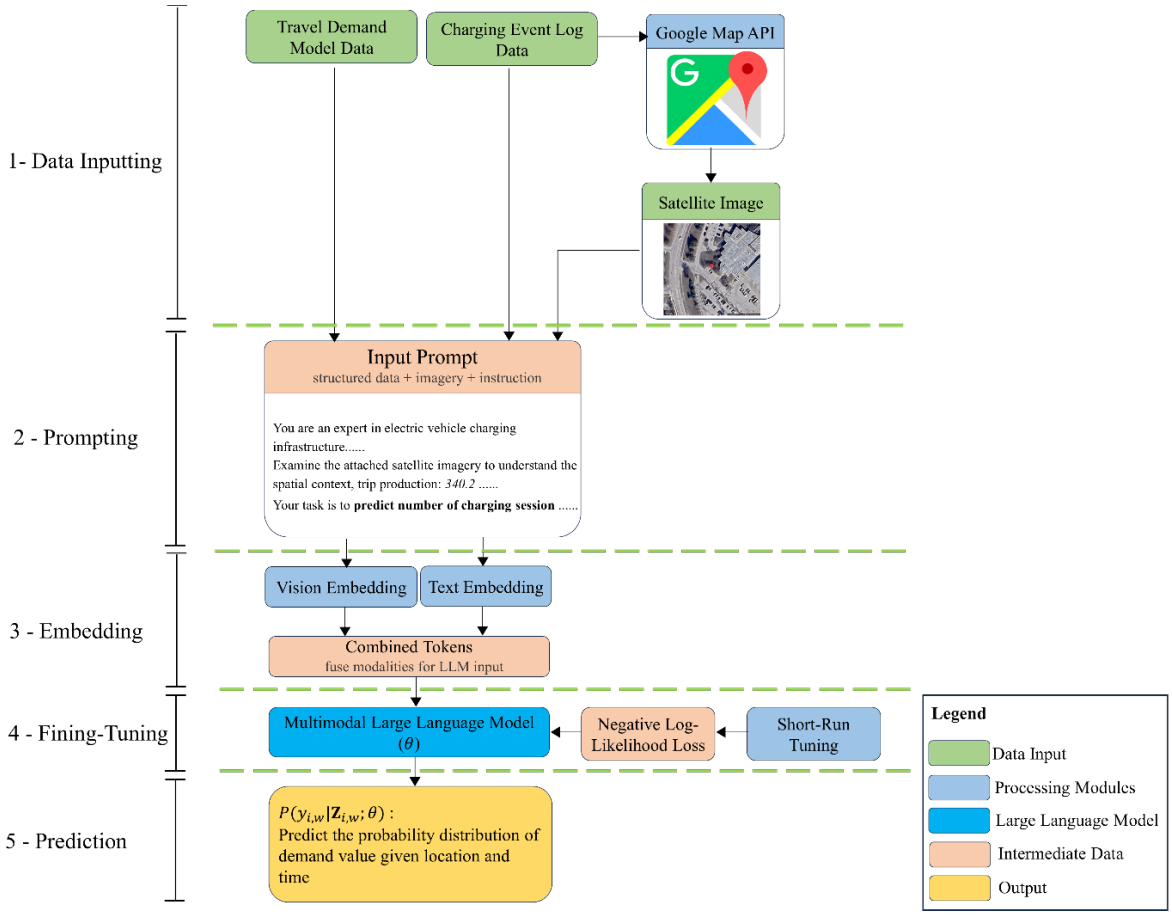


Figure 2 The workflow chart of VisionEV

In the prompting stage, the collected structured metadata—including travel demand indicators, and seasonal factors—is converted into a natural-language prompt using a task-specific template. This textual encoding aligns the structured data with the semantic representations learned by pretrained LLMs, facilitating deeper contextual reasoning. Subsequently, during embedding, each charging station is represented as a multimodal prompt-image pair. Satellite images are processed by PaliGemma's built-in SigLIP vision tower (frozen), producing patch-level visual tokens. Simultaneously, the structured prompts are tokenized by a SentencePiece tokenizer and encoded using a pretrained T5-style text encoder. A learned linear projection layer aligns these visual embeddings with the textual embeddings into a shared latent space. The combined multimodal sequence is then passed into a decoder-only transformer model for integrated spatial-semantic interpretation.

Next, fine-tuning adapts the pretrained multimodal model to the specific characteristics of EV charging demand. The model is trained using real-world data and optimized via a regression loss function (such as root mean squared error), employing cosine learning rate scheduling for stable and effective convergence. This stage ensures the model captures detailed spatial patterns, behavioral dynamics, and temporal variabilities inherent in EV charging data. Finally, at the demand prediction stage, VisionEV leverages its trained multimodal representations to predict charging station usage. At inference, new satellite images and structured prompts are paired and processed by the trained model, generating accurate forecasts even for stations with limited or no historical data. The resulting model integrates both visual spatial understanding and structured contextual reasoning, enabling robust and generalizable predictions across diverse urban and planning scenarios.

The proposed VisionEV framework is city-agnostic by design. It requires only globally accessible inputs—public satellite imagery (e.g., Google/Bing/OSM sources), region-level traffic indicators (e.g., AADT, trip production or their proxies), and temporal descriptors—which are commonly available across metropolitan areas. The multimodal backbone is pretrained on large-scale, diverse image–text corpora, providing strong prior knowledge for cross-domain transfer. In addition, the lightweight fine-tuning strategy allows rapid adaptation to new regions with limited local samples.

The following sections detail the key components of the VisionEV methodology: prompt design (Section 3.2), demand forecasting formulation (Section 3.3), and fine-tuning strategy (Section 3.4).

3.2 Prompt Design

The prompt serves as the central mechanism for converting diverse spatial and structured inputs into a format compatible with language models. Our carefully designed prompts embed explicit spatial attributes, simulate expert reasoning, and incorporate perceptual context within a unified generative framework.

3.2.1 Expert Role Conditioning

To effectively leverage the reasoning capabilities of the model, each prompt begins with a clear, role-based instruction that positions the LLM as an expert EV infrastructure analyst. This zero-shot conditioning strategy enhances the alignment between general-purpose language models and specialized domain forecasting tasks, promoting structured and contextually appropriate model responses (Shanahan et al. 2023).

3.2.2 Natural Language Encoding of Planning Metadata

Structured planning attributes—including AADT, trip production, and temporal indicators are encoded into the prompt using natural language. Converting numerical and categorical metadata into descriptive sentences allows alignment with the semantic space learned during model pretraining, reducing the reliance on extensive manual feature engineering. Representative examples of this encoding include:

- "The station is situated in a traffic assignment zone with production value of 340.22."
- "Current nearby traffic flow (AADT) measures 4079.97 vehicles."
- "Time period: Fall season, week 44 of the year."

Incorporating these context-rich descriptions ensures that the LLM effectively captures critical spatial and temporal factors influencing EV charging behavior.

3.2.3 Integration of Satellite Imagery as Spatial Context

A distinguishing innovation in our approach is the incorporation of satellite imagery to provide perceptual spatial context. Each satellite image is processed through a SigLIP vision encoder, extracting semantic visual cues such as land use classifications, building density, and road network layouts. These visual embeddings are then linearly projected and merged with the tokenized textual prompt. This integration enables the LLM to simultaneously reason over structured metadata and unstructured visual information, effectively capturing spatial elements such as parking availability, proximity to commercial centers, and overall accessibility—features typically challenging to encode manually.

3.2.4 Instruction-Constrained Output

To guide the model explicitly toward numerical forecasting, each prompt concludes with a clear instruction requesting a singular numeric prediction:

"Your task is to predict the number of charging sessions for this station during the specified week hour. In your answer, you should provide the value of your prediction only."

In summary, our prompt design strategically combines explicit role-based instructions, descriptive encoding of structured metadata, and innovative integration of satellite imagery to create a comprehensive multimodal input. This design facilitates enhanced extraction of both explicit spatial attributes and implicit environmental cues, ultimately leading to improved accuracy and reliability in EV charging demand predictions. A complete example of such a prompt is provided below:

INPUT
You are an expert in electric vehicle charging infrastructure,
who is good at **<charging demand forecasting>**.
We are currently assessing a location in *Kansas City, Missouri*.
Please review the following station data:

- Examine the attached satellite imagery to understand the spatial context
- The station is situated in a traffic assignment zone with trip production value of *340.22*
- Current nearby traffic flow (AADT) measures *4079.97* vehicles
- Time period: *Fall season, week 44 of the year*
  Now, pay attention! Your task is to **<predict the number of charging sessions for this station during the specified week >**
  In your answer, you should provide the value of your prediction only.

TARGET
10

3.3 Demand Forecasting Formulation

The prediction of EV charging demand requires the effective integration of diverse data sources, including spatial information from satellite imagery and structured planning. Formally, the forecasting problem considers a dataset comprising observations from EV charging stations, defined as Eq. (1):

$$D = \left\{\left(\mathbf{I}_i, \mathbf{s}_{i,w}, d_{i,w}\right)\right\}_{i=1}^{N} \quad \text{Eq. (1)}$$

where each data instance consists of:

- A satellite image $\mathbf{I}_i \in \mathbb{R}^{H\times W\times C}$, representing the station's spatial surroundings, with dimensions of height H, width W, and C color channels.

- A structured planning vector $\boldsymbol{s}_{i,w} \in \mathbb{R}^d$, containing relevant urban and planning-related variables, including AADT, and seasonal context at station i in week w.
- The observed charging demand $d_{i,w} \in \mathbb{Z}_{\geq 0}$, representing weekly EV charging sessions at station i in week w.

Our objective is to build a multimodal predictive model, formally described as Eq. (2):

$$f: (\mathbf{I}_i, \mathbf{s}_{i,w}) \rightarrow \hat{d}_{i,w} \qquad \text{Eq. (2)}$$

where $\hat{d}_{i,w}$ is the predicted weekly EV charging demand. As shown in Figure 3, VisionEV addresses this forecasting task through a structured multimodal framework. First, satellite images and structured data are converted into a multimodal input sequence via embedding. These combined multimodal tokens are then processed through a decoder-only transformer model, effectively merging visual and textual contexts. Finally, the transformer generates discrete tokens, which are decoded to yield explicit numerical forecasts of weekly charging sessions. We adopt this tokenization–decoding–detokenization formulation because decoding over a restricted numeric vocabulary naturally yields non-negative integer outputs, avoiding ad-hoc clipping or rounding, and because a generative head remains faithful to PaliGemma's next-token training objective, preserving its pretrained cross-modal alignment and in-context generalization—benefits that are especially valuable in few-/zero-shot settings. The detokenization step is a lightweight, parameter-free numeric parse, keeping the pipeline simple while ensuring constraint-consistent, promptable predictions.

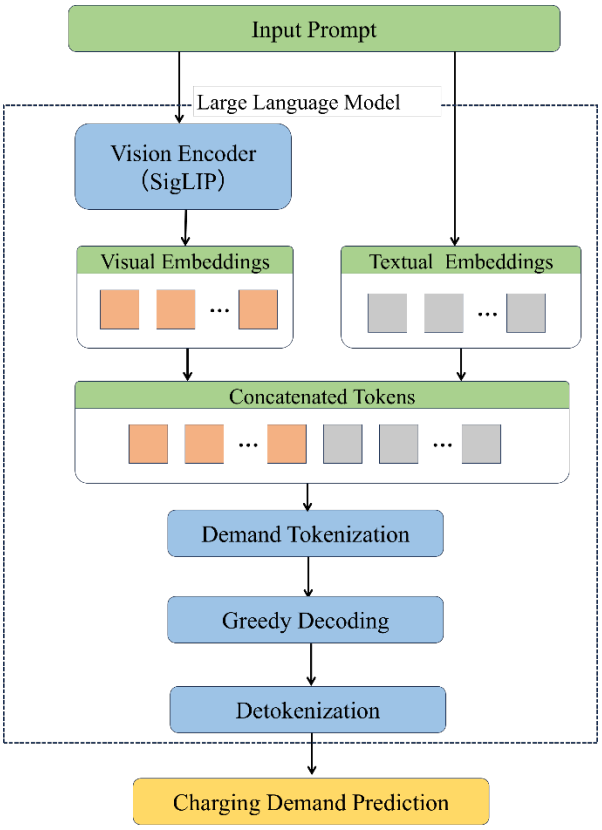


Figure 3 VisionEV demand prediction framework

3.3.1 Multimodal Embedding and Unified Input Sequence

VisionEV initially encodes multimodal inputs into meaningful spatial-textual representations.

Visual Embedding

Satellite images $\mathbf{I}_i$ are processed through the frozen, built-in SigLIP encoder of PaliGemma, denoted $\phi_v(\cdot)$, yielding visual patch embeddings (as Eq. (3)):

$$\mathbf{v}_i = \phi_v(\mathbf{I}_i) \in \mathbb{R}^{n_v \times \mathrm{d}_v} \quad \text{Eq. (3)}$$

where n_v represents the number of visual patches, and d_v their embedding dimensionality. These visual embeddings are projected into the language-compatible token embedding space using a linear transformation shown by Eq. (4):

$$\mathbf{z}_i^{(img)} = \mathbf{W}_v \mathbf{v}_i, \quad \mathbf{W}_v \in \mathbb{R}^{d' \times \mathrm{d}_v} \quad \text{Eq. (4)}$$

with embedding dimension d' matching the language model's input requirement. VisionEV's ability to seamlessly integrate these visual tokens into the language domain significantly enhances spatial context understanding.

Textual Embedding

Structured attributes $\mathbf{s}_{i,w}$ are translated into descriptive natural language prompts, e.g., *"This station has an AADT of 8500 and season is fall."* These textual prompts $\mathbf{T}_{i,w}$ are tokenized and embedded via a pretrained textual encoder $\phi_t(\cdot)$ as Eq. (5):

$$\mathbf{z}_{i,w}^{(txt)} = \phi_t(\mathbf{T}_{i,w}) \in \mathbb{R}^{n_t \times d'} \quad \text{Eq. (5)}$$

where n_t is the number of textual tokens, and d matches the visual embedding dimension.

Unified Multimodal Sequence Representation

The visual and textual embeddings are concatenated to form a unified multimodal sequence as Eq. (6):

$$\mathbf{Z}_{i,w} = [\mathbf{z}_i^{(img)}; \mathbf{z}_{i,w}^{(txt)}] \in \mathbb{R}^{(n_v+n_t) \times d'} \quad \text{Eq. (6)}$$

This multimodal embedding strategy enables large language model's transformer-based decoder to simultaneously reason over spatial imagery and structured textual data, thereby effectively capturing complex spatial-contextual relationships inherent to EV charging demand prediction tasks.

3.3.2 Token-Level Generation and Greedy Decoding

We frame the EV charging demand prediction as a discrete token-generation task, aligning naturally with the backbone large language model's decoder-based language generation architecture. This token-based approach exploits multimodal token generation capability, which inherently supports a direct mapping between the model's output and discrete charging session counts.

Demand Tokenization

We convert the continuous demand counts $d_{i,w}$ into discrete tokens from a predefined integer-valued vocabulary $V \subset \mathbb{Z}_{\geq 0}$ as shown by Eq. (7):

$$y_{i,w} = \mathrm{Tok}(d_{i,w}) \quad \text{Eq. (7)}$$

Given the multimodal input sequence $\mathbf{Z}_{i,w}$, a probability distribution over tokens is generated as Eq. (8):

$$P(y_{i,w} | \mathbf{Z}_{i,w}; \theta) = \mathrm{softmax}(f_\theta(\mathbf{Z}_{i,w})) \quad \text{Eq. (8)}$$

where $f_\theta(\cdot)$ represents the output logits of the large language model, parameterized by θ.

Greedy Decoding

At inference, we employ greedy decoding, selecting the token with the highest predicted conditional probability as Eq. (9):

$$\hat{y}_{i,w} = \arg\max_{y \in V} P(y_{i,w} | \mathbf{Z}_{i,w}; \theta) \quad \text{Eq. (9)}$$

Greedy decoding is justified mathematically by the single-token nature of the EV demand prediction task, as it maximizes the likelihood of the single most probable demand token, ensuring both optimality and computational efficiency in practical scenarios.

Decoder Architecture and Parameter Configuration

The output generator in VisionEV is a Transformer decoder-only module from the Gemma-3B family configured for single-step, causal generation of a numeric token. Each layer follows a pre-norm layout—RMSNorm → Multi-Head Self-Attention (causal mask) → RMSNorm → Feed-Forward (GELU)—which improves stability when fine-tuning on short sequences with heterogeneous multimodal inputs. The decoder's model width is matched to the multimodal embedding space so that the concatenated visual–text tokens (Eq. (6)) can be consumed without additional adapters. We employ Rotary Position Embeddings (RoPE) for position encoding to preserve relative ordering under variable prompt lengths, and set head size and FFN expansion to standard Gemma ratios for a balanced capacity–latency trade-off. The LM head projects the final hidden state to a compact integer-token vocabulary used for weekly sessions; weights are untied from input embeddings to avoid capacity bottlenecks for the numeric head. Key decoder hyperparameters are summarized in Table 1.

Table 1 Principal decoder hyperparameters

Component	**Setting / Symbol**	**Description**
Architecture	Transformer (decoder-only)	Autoregressive, causal self-attention
# Layers	24	Depth of decoder stack
Hidden dim	$\mathrm{d}' = 2048$	Token & attention embedding size
Attention heads	16	Head dim = 128
FFN dim	8192	$4 \times \mathrm{d}'$; GELU activation
Normalization	RMSNorm	Pre-norm per sublayer
Positional encoding	RoPE	Rotary position embeddings
Dropout	0.10	Attention & FFN
Max sequence length	256	Prompt tokens + single numeric token

Detokenization and Output Mapping

The selected token is mapped back into the integer-valued demand domain via a deterministic detokenization function as Eq. (10):

$$\hat{d}_{i,w} = \mathrm{DeTok}(\hat{y}_{i,w}) \quad \text{Eq. (10)}$$

to ensure direct usability of model predictions.

3.3.3 Learning Objective

The VisionEV model parameters θ are trained by minimizing the Negative Log-Likelihood (NLL) loss over the training dataset in Eq. (11):

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} logP(y_{i,w}|\mathbf{Z}_{i,w};\theta) \quad \text{Eq. (11)}$$

This learning objective explicitly aligns the model parameters with the goal of maximizing the probability assigned to the correct discrete demand tokens, thus directly optimizing the accuracy of EV charging demand forecasts.

VisionEV formulates EV charging demand prediction as a multimodal token generation problem, leveraging its unique capability to seamlessly integrate visual spatial information and structured urban context. This unified token embedding and generation framework enables effective reasoning across heterogeneous data types, capturing the spatial-contextual complexities inherent in EV demand forecasting. Compared to conventional regression methods, VisionEV offers superior interpretability and robustness, making it well-suited for infrastructure planning.

3.4 Tuning Method

Fine-tuning pretrained models is a widely adopted strategy to tailor general-purpose models to specific tasks, particularly when the target domain, such as EV charging demand forecasting, displays significant spatial heterogeneity. VisionEV employs a structured, lightweight fine-tuning approach explicitly designed to efficiently adapt multimodal pretrained representations to the unique spatial-contextual features of EV infrastructure planning.

Instead of initializing training from scratch—which is computationally costly and prone to overfitting—VisionEV utilizes pretrained model weights from PaliGemma. Specifically, the pretrained vision encoder (SigLIP) parameters are kept frozen to retain their robust visual representations. In contrast, the linear projection matrix $\mathbf{W}_v$ (see Eq. (4)), textual embedding layers $\phi_t(\cdot)$, and transformer decoder parameters θ are fine-tuned, allowing targeted adaptation to EV-specific forecasting tasks.

This fine-tuning strategy is crucial for VisionEV's performance across varying data scenarios. In few-shot settings, where only a limited subset of training data (e.g., 5%–20%) is available, short-run fine-tuning rapidly aligns pretrained multimodal representations with EV-specific patterns, allowing VisionEV to effectively capture essential spatial-contextual insights from sparse datasets. Conversely, in zero-shot settings, VisionEV relies heavily on the generalization capability of its pretrained visual and textual embeddings. By maintaining the frozen visual encoder and leveraging semantic priors from pretrained layers, VisionEV can accurately forecast demand at entirely unseen locations without local data, demonstrating strong generalization across diverse spatial contexts.

Implementation details of the optimization process—learning-rate schedule, optimizer updates, periodic validation, and pseudocode —are moved to Appendix A for clarity, allowing this section to focus on the core fine-tuning design.

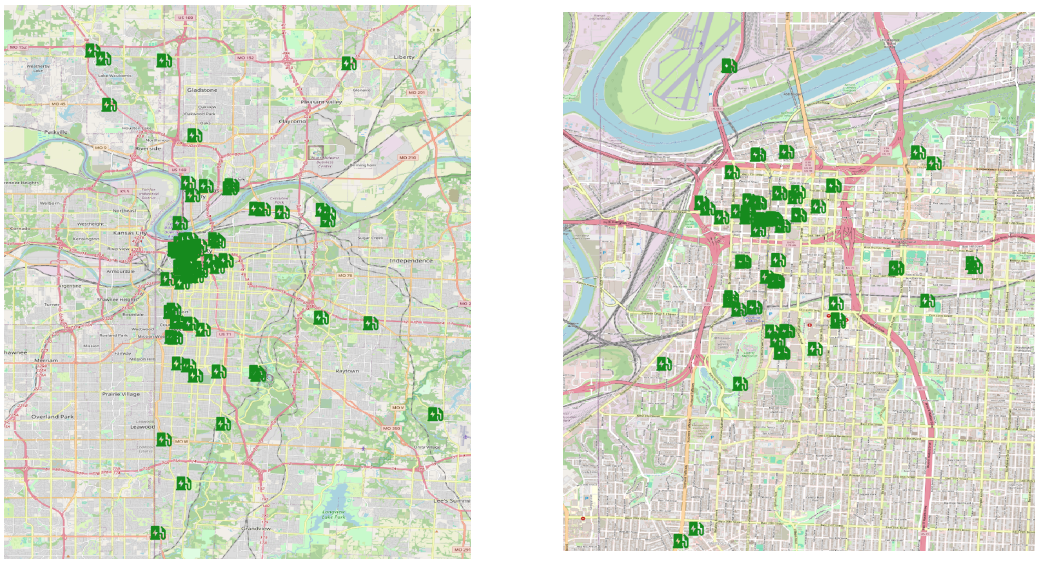
4. Case study

To evaluate the effectiveness of VisionEV, our multimodal EV charging demand forecasting framework, we conduct an extensive empirical study using real-world data from Kansas City, Missouri (KCMO). This section details the dataset composition, environment configuration, and analytical methods employed to rigorously benchmark VisionEV against established predictive models. We evaluate predictive accuracy in various experimental setups and conduct targeted ablation studies to quantify the specific contributions of textual prompting, satellite imagery integration, and the fine-tuning strategy.

4.1 Data Description

The dataset used in this study integrates three heterogeneous yet complementary sources of information: charging event logs, travel demand model outputs, and satellite imagery. All datasets are geo-aligned at the station level and cover the same spatial and temporal domain, enabling a consistent modeling framework. These datasets were originally curated for our prior work on EV driver range anxiety analysis (Song and Hu 2021) and are repurposed here to evaluate forecasting performance.

Charging Event Log Data: We begin with a charging event log dataset covering 189 public EV charging stations in KCMO. As shown in Figure 4, these stations span both downtown and suburban areas, capturing a wide range of land uses and spatial contexts. The logs contain detailed information for 190,116 charging events collected between 2021 and 2023, including start/end timestamps, and station coordinates. We aggregate these records into a weekly time resolution, resulting in a target variable: the number of charging sessions per station per week. This aggregation reflects real-world operational considerations and aligns with the temporal resolution used by regional planners.



(a) Overview of KCMO (b) Downtown region with higher station density

Figure 4 The spatial distribution of charging stations in the dataset

Travel Demand Model Data: To incorporate region-specific transportation activity, we extract planning features from the regional travel demand model provided by the Mid-America Regional Council (MARC). The model covers 2,510 traffic analysis zones (TAZs) and includes origin–destination matrices, trip production statistics, and roadway metrics.

For each charging station, we extract: (1) Trip production from the containing TAZ, representing the number of trips originating in that zone. (2) Annual Average Daily Traffic on adjacent road segments, reflecting potential vehicle exposure. These structured features encode both macroscopic travel behavior and local traffic intensity, which are important contextual drivers of EV charging demand.

Satellite Imagery Data: A key innovation in our approach is the use of satellite imagery to represent unstructured spatial context. Using the Google Static Maps API, we retrieve high-resolution images (1640 × 1640 pixels, zoom level 19) centered on each charging station's coordinates. These images provide consistent, and visual coverage of the site's environment—including buildings, roads, vegetation, and land use heterogeneity.

To illustrate the diversity captured by these images, Figure 5 presents four sampled stations: Figure 5-(a) shows a downtown commercial building, typically associated with public or workplace charging; Figure 5-(b) depicts a residential street, reflecting likely home-based charging activity; Figure 5-(c) is an industrial

site near a railyard, indicative of fleet or logistic-oriented usage; Figure 5-(d) shows a recreational area near a zoo, aligned with visitor or tourist-based demand.

These examples highlight the rich semantic variance captured through visual context, which cannot be easily quantified through tabular descriptors alone. The use of imagery allows the model to infer patterns such as land use type, accessibility, and density—factors that are strongly linked to charging behavior but often missing from conventional features.



(a) (b)

(c) (d)

Figure 5 The Sampled Satellite Image

From these sources, we construct a structured dataset of 28,566 station-week samples, each including the target charging demand and the following features: satellite image, trip production, AADT, a seasonal indicator (Winter, Spring, Summer, or Fall) and week of year.

The dataset is randomly partitioned into three subsets for model training and evaluation:

- Training set: 22,852 samples (80%)
- Validation set: 2,856 samples (10%)
- Testing set: 2,858 samples (10%)

This multimodal, spatially diverse dataset provides a comprehensive testbed for benchmarking multimodal forecasting methods. Our proposed model is designed to leverage this multimodality, learning from both structured attributes and perceptual spatial signals to produce accurate demand forecasts.

4.2 Environment Setup

All experiments are conducted in a controlled Linux-based environment using Google Colab Pro, equipped with a single NVIDIA L4 GPU (24 GB VRAM). This hardware setting supports efficient fine-tuning of large-scale multimodal LLMs while remaining accessible and reproducible for academic and prototype use cases. The training pipeline is implemented using JAX, a high-performance numerical computing library, alongside the BigVision framework, which is optimized for large multimodal model training and evaluation.

VisionEV is fine-tuned from the publicly available PaliGemma-3B checkpoint, which includes a SigLIP vision encoder and a Gemma-based text decoder. This model is selected for its strong performance in multimodal reasoning tasks and its compatibility with both structured and unstructured inputs. Optimization is performed using stochastic gradient descent (SGD) with a cosine learning rate schedule. The initial learning rate is set to 0.005, and a linear warm-up strategy is applied during the first 10% of the training steps to stabilize early convergence.

Training is conducted for 200 epochs with a batch size of 64, and model performance is evaluated every 10 training steps using a held-out validation set. All satellite images are preprocessed to match the model's pretraining configuration—specifically, resized to 224 × 224 pixels and normalized to the same mean and standard deviation values used during initial SigLIP training. This ensures alignment between pretraining and downstream fine-tuning, which is critical for preserving generalization performance.

Text inputs are tokenized using the SentencePiece tokenizer built into the backbone large language model. Each input sequence follows a structured prefix–suffix format, where the prefix embeds contextual information such as station planning features, while the suffix represents the target prediction. We use a maximum sequence length of 256 tokens, applying full attention to the prefix (i.e., encoder-style) and causal attention to the suffix (i.e., decoder-style). This configuration mirrors typical encoder-decoder training schemes in language modeling. The model is optimized using the negative log-likelihood (NLL) loss, computed over the suffix tokens, which correspond to the predicted weekly charging demand as a discrete numerical token.

Model performance is assessed using two standard evaluation metrics commonly applied in time-series regression tasks:

- Root Mean Squared Error: Sensitive to large prediction errors and reflects general accuracy.
- Mean Absolute Error: Reflects average deviation and is robust to outliers.

Hyperparameters—including batch size, sequence length, and learning rate schedule—are selected through empirical tuning on the validation set. All baseline models (e.g., Linear Regression, LightGBM, GCN, and Gemma) are trained using standardized settings recommended by their original implementations to ensure a fair and consistent comparison across models. This setup enables robust benchmarking and highlights the advantages of our proposed multimodal LLM-based forecasting approach in terms of both predictive accuracy and architectural flexibility.

4.3 Performance evaluation

To systematically assess VisionEV's predictive accuracy, robustness, and generalization capability, we conduct comprehensive performance comparisons against established baseline models across three clearly defined forecasting scenarios. These scenarios are explicitly structured to showcase VisionEV's multimodal strengths and its practical applicability in diverse real-world planning contexts:

1. **Full-shot Prediction:** Evaluates the maximum predictive accuracy achievable when complete training data is available for all 189 EV charging stations.
2. **Few-shot Prediction:** Measures the model's efficiency and adaptability by limiting training data availability to only 5%, 10%, 15%, and 20%, simulating scenarios common in early deployments or regions with sparse historical data.
3. **Zero-shot Prediction on Within-City Unseen Stations:** Tests the generalization ability of the model by training on random subsets (20%, 40%, 60%, 80%) of the 189 stations and testing on the held-out remainder—an evaluation crucial for deployment in new or currently unmonitored areas.

We select baselines spanning complementary modeling paradigms to offer robust comparative benchmarks. To ensure comparability, all non-vision baselines are trained on the same structured inputs used by VisionEV—namely AADT, trip production, and temporal indicators (season $\in$ {Winter, Spring, Summer, Fall} and week-of-year). In addition, to supply spatial context for models that do not consume images, we append land-use one-hot indicators at the station location: Institutional (LDU1), Transportation (LDU2), Commercial (LDU3), Residential (LDU4), Recreational (LDU5), Vacant (LDU6), and Industrial (LDU7). These binaries are derived from parcel land-use labels at the station point and concatenated to the tabular feature vector. Continuous features are standardized (zero mean, unit variance) on the training split; binary indicators remain in {0,1}.

By design, this gives classical/graph/text baselines access to the same planning variables as VisionEV and, via LDU1–LDU7, a tabular proxy for spatial context, whereas VisionEV learns spatial semantics directly from satellite imagery and therefore does not use handcrafted land-use indicators.

- **Linear Regression (LR):** Minimal-complexity reference to quantify gains from more sophisticated learners; trained with 5-fold cross-validation on the training split for regularization.
- **LightGBM:** Strong tree-based benchmark for structured forecasting; trained with 5-fold cross-validation to tune tree depth, learning rate, and regularization.
- **Graph Convolutional Network (GCN):** Captures inter-station dependence via a relational graph; node features are the same tabular vector as above, and edges encode spatial proximity (e.g., k-NN over geodesic distances computed from station coordinates).
- **Gemma (text-only LLM):** To ensure an equal comparison, we use a prompt that mirrors VisionEV's template while removing the instruction to examine satellite imagery and explicitly stating land use in natural language (e.g., "land use is Commercial"). The structured attributes—AADT, trip production, season, and week-of-year—are verbalized, and the model is instructed to output only the numeric prediction with no image tokens are provided.

4.3.1 Full-shot Prediction

In the full-shot setting, all models are trained with access to complete data across all traffic zones, allowing us to assess their maximum predictive capability under ideal conditions. As reported in Table 2, our proposed model, VisionEV, achieves the best performance among all evaluated methods, with an RMSE of 2.87 and MAE of 1.98. Compared to the strongest baseline, LightGBM (RMSE: 2.90, MAE: 2.09), VisionEV yields a relative improvement of 1.0% in RMSE and 5.3% in MAE. Although the numerical gains may appear modest, they are meaningful in high-variance, real-world deployment scenarios, where

marginal improvements in predictive accuracy can lead to substantial enhancements in infrastructure efficiency and planning decisions. Additionally, VisionEV favors portability (no handcrafted spatial features), constraint-consistent outputs, and promptable adaptation to new planning assumptions—trade-offs that support robust deployment across geographies with minimal re-engineering.

Among the baselines, LightGBM clearly outperforms Linear Regression, illustrating the value of nonlinear modeling in capturing complex relationships within structured demand data. The GCN baseline benefits from spatial structure by modeling connectivity among charging stations but still falls behind VisionEV. This is likely due to its limited capacity to interpret the rich contextual information embedded in the built environment—such as land use patterns or access routes—that are not easily captured through topological graphs alone.

The Gemma model, which relies solely on prompt-based learning over textual metadata, performs better than both LR and GCN by leveraging the semantic richness of pretrained LLMs. However, its lack of visual input constrains its ability to fully understand the physical context that often drives real-world charging behavior. Without visual features, the model struggles to generalize across zones with varying urban morphology and land use characteristics.

By contrast, VisionEV's multimodal architecture effectively integrates both structured textual prompts and high-resolution satellite imagery. The visual modality provides spatial cues—such as street density, green space, or commercial activity—while the textual input encodes policy-relevant features like average daily traffic, and seasonal factor. This joint reasoning over visual and semantic features enables VisionEV to capture both fine-grained environmental cues and high-level planning knowledge, resulting in consistently superior predictions.

Overall, these results demonstrate the advantage of combining vision and language in EV demand forecasting and establish VisionEV as a new state-of-the-art under full-shot conditions.

Table 2 Performance comparison on full-shot prediction

Metric	**RMSE**	**MAE**
Linear Regression	4.91	3.60
LightGBM	2.90	2.09
GCN	3.77	2.79
Gemma (text-only LLM)	3.18	2.50
VisionEV	**2.87**	**1.98**

4.3.2 Few-shot Prediction

To evaluate the sample efficiency and adaptability of our proposed multimodal framework under data-limited conditions, we conduct a series of few-shot experiments by randomly selecting 5%, 10%, 15%, and 20% of the available training data. This experimental setup simulates real-world situations where labeled data may be scarce—such as newly developed regions, pilot deployments, or datasets fragmented due to privacy or infrastructure constraints. The results, shown in Table 3, reveal clear differences in model robustness under limited data conditions. Traditional models like Linear Regression struggle considerably, reaching an average RMSE of 10.98 and MAE of 8.33, underscoring their limitations in capturing nonlinear relationships with sparse data. While GCN benefits from spatial connectivity, it still suffers under low-data conditions (RMSE: 9.64, MAE: 7.60), highlighting its reliance on abundant labeled examples to extract meaningful spatial patterns.

Among the baseline models, LightGBM exhibits stronger robustness with an average RMSE of 3.53 and MAE of 2.39, thanks to its gradient-boosting design and effective handling of structured features. Gemma,

the text-only LLM model that relies on natural language prompts from metadata, performs similarly (RMSE: 3.78, MAE: 2.86) and benefits from pretrained language knowledge. However, the best performance is achieved by VisionEV, which consistently outperforms all baselines across all data levels. With an average RMSE of 3.30 and MAE of 2.29, VisionEV demonstrates remarkable learning efficiency, with clear gains as data availability increases—from RMSE 3.53 at 5% data to 3.14 at 20%, and MAE improving from 2.40 to 2.20. These results underscore the advantage of leveraging multimodal context—where visual inputs capture spatial semantics and textual prompts encode structured planning variables—enabling VisionEV to learn quickly and generalize effectively even in severely data-sparse environments.

Table 3 Performance comparison on few-shot prediction

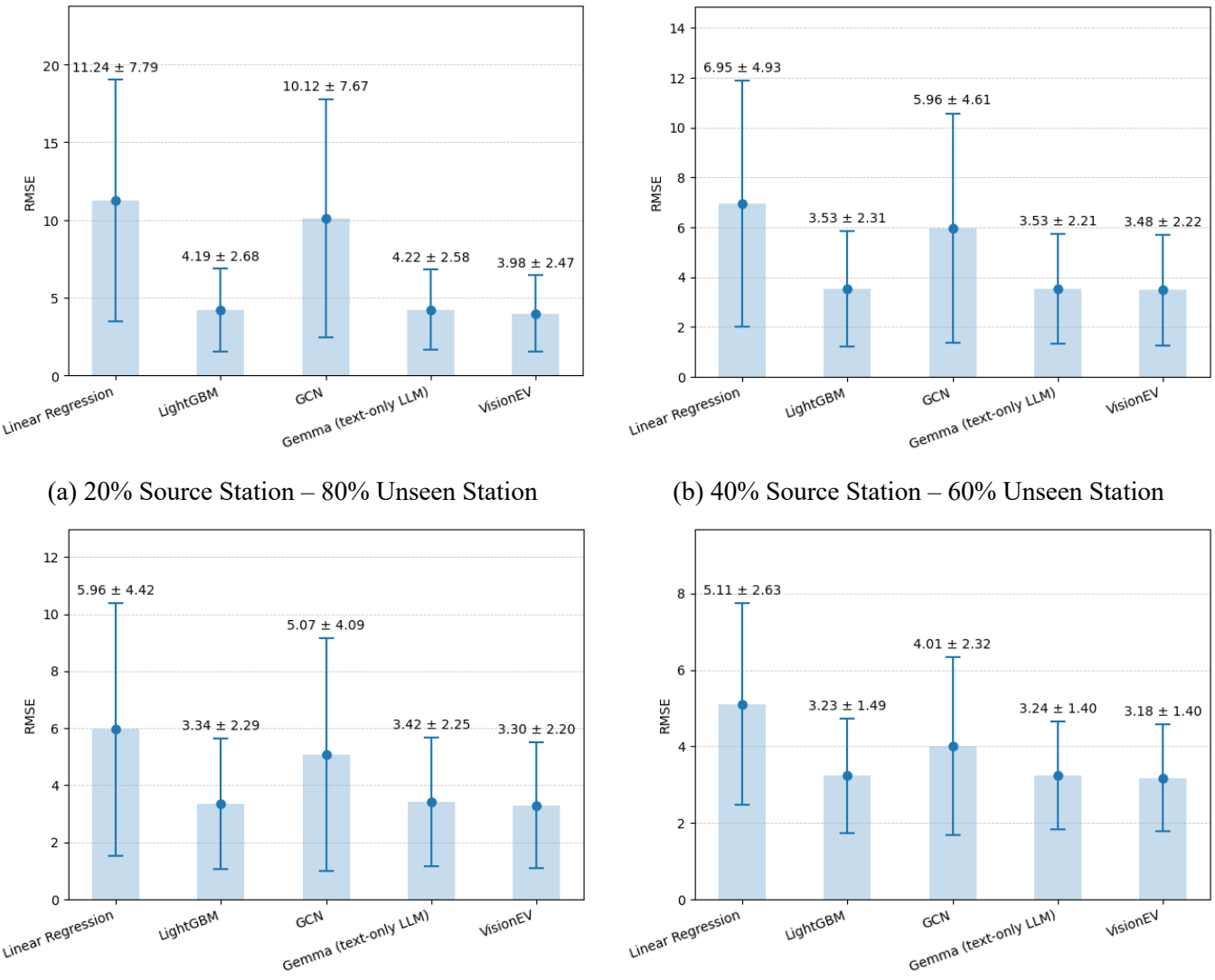
Metric	**RMSE**					**MAE**				
Training data percent	5%	10%	15%	20%	Mean	5%	10%	15%	20%	Mean
Linear Regression	11.65	11.28	10.46	10.53	10.98	8.39	8.35	8.26	8.31	8.33
LightGBM	3.84	3.45	3.42	3.39	3.53	2.61	2.33	2.31	2.29	2.39
GCN	9.63	9.71	9.63	9.58	9.64	7.70	7.74	7.63	7.33	7.60
Gemma (text-only LLM)	3.91	3.76	3.72	3.71	3.78	3.04	2.80	2.81	2.79	2.86
VisionEV	**3.53**	**3.33**	**3.19**	**3.14**	**3.30**	**2.40**	**2.29**	**2.28**	**2.20**	**2.29**

4.3.3 Zero-shot Prediction on Within-City Unseen Stations

Beyond few-shot learning, a critical test of model robustness is its ability to perform zero-shot forecasting—that is, making accurate predictions in entirely unseen charging stations without any fine-tuning on those targets. This setting is especially relevant for practical deployment, where charging demand predictors must generalize to newly installed or data-scarce stations. To evaluate this, we conduct cross-station adaptation experiments by randomly and repeatedly selecting 20%, 40%, 60%, and 80% of the 189 public EV charging stations as source domains for training, while using the remaining stations as target domains for testing. No training samples from the target stations are used during training or adaptation, making this a stringent test of out-of-distribution generalization. In this setting, training and test sets are spatially disjoint within the same city, ensuring that the model never observes the test stations' imagery or local built environment during training. This protocol removes station-level overlap and temporal leakage and probes whether the model can interpret novel spatial scenes. While data remain within one metro area, this serves as a small-scale form of domain generalization to spatially unseen zones.

As the mean and standard deviation of RMSE summarized in Figure 6, VisionEV consistently achieves the best performance across all source-target splits, demonstrating superior zero-shot generalization. When trained on only 20% of the stations, VisionEV attains a mean RMSE of 3.98, outperforming LightGBM (4.19), Gemma (4.22), and GCN (10.12), while also showing the lowest standard deviation (2.47) among all models. As the proportion of source stations increases to 40%, 60%, and 80%, VisionEV maintains its lead, with RMSEs of 3.48, 3.30, and 3.18, respectively, and continues to yield more stable predictions than all baselines. In contrast, GCN exhibits severe degradation, with RMSE values consistently above 4.0, highlighting its difficulty in transferring learned spatial dependencies without station-specific training data. Linear Regression, although simpler, shows high variance and limited accuracy, failing to capture complex demand patterns across heterogeneous urban contexts.

These results underscore the power of VisionEV's multimodal architecture, which integrates satellite imagery and structured textual prompts to form a rich, generalizable representation of charging station context. While baseline models rely on either tabular learning (LightGBM), spatial graphs (GCN), or language-only prompts (Gemma), VisionEV's ability to leverage both visual semantics (e.g., surrounding land use, street layout) and domain-specific metadata (e.g., AADT, season) allows it to reason effectively about unseen stations. This makes VisionEV especially suitable for real-world EV infrastructure planning, where predictions must often be made in locations with little or no historical data. Although the zero-shot experiments are conducted within Kansas City, they effectively serve as a miniature form of cross-region generalization, where the model predicts charging demand for spatially unseen zones. The consistent performance in these settings provides preliminary evidence that VisionEV can generalize beyond its training distribution to new urban contexts.



(a) 20% Source Station – 80% Unseen Station

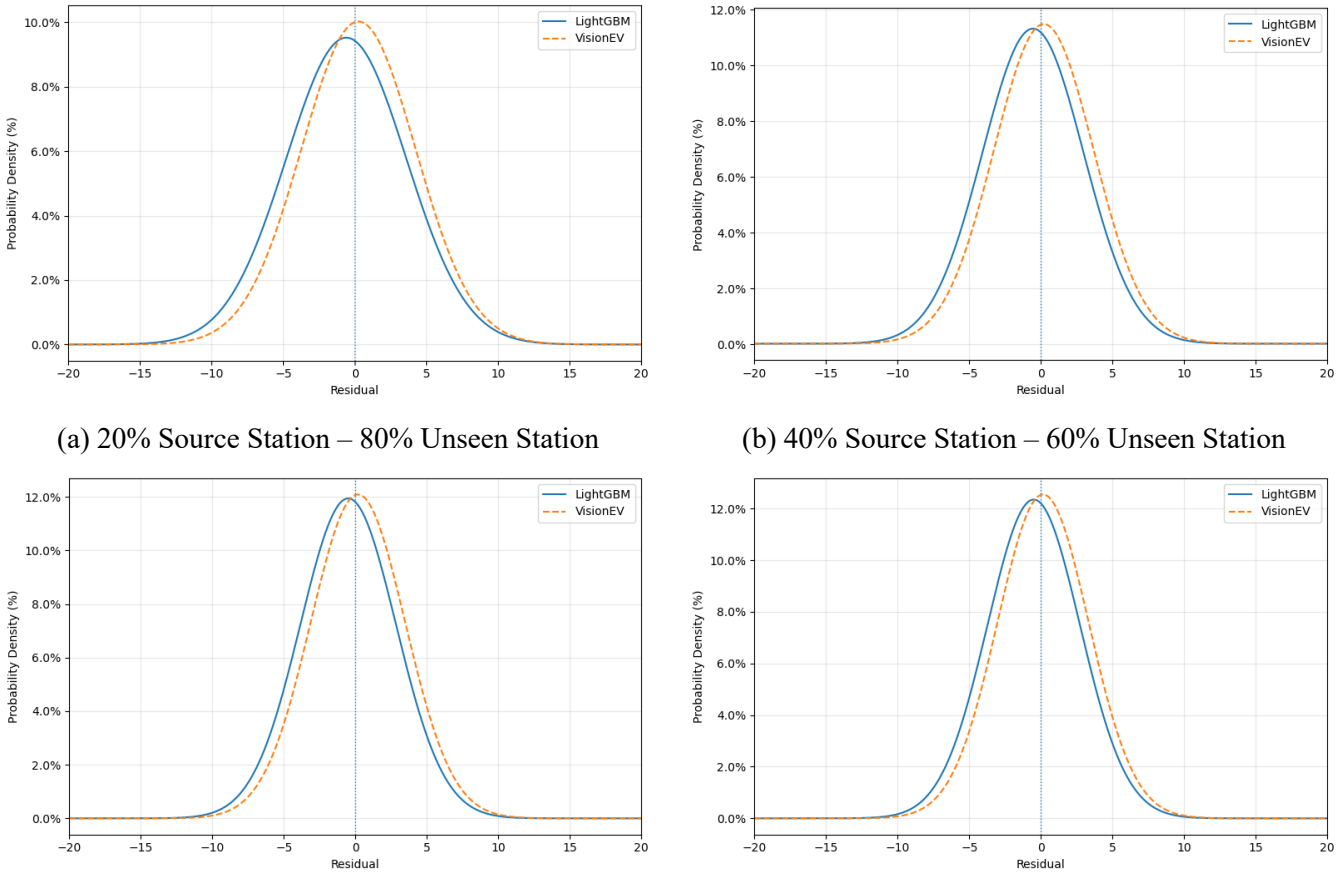
(b) 40% Source Station – 60% Unseen Station

(c) 60% Source Station – 40% Unseen Station

(d) 80% Source Station – 20% Unseen Station

Figure 6 RMSE comparison for zero-shot prediction on within-city unseen stations

For further analysis, we plot Figure 7, which compares the residual distributions of VisionEV and the best baseline (LightGBM) under the zero-shot setting. Across all source–target splits (20/80, 40/60, 60/40, 80/20), VisionEV's residuals are more tightly centered on zero and exhibit greater symmetry, whereas LightGBM shows a slight right skew. This pattern suggests that VisionEV reduces systematic bias—mitigating both over- and under-estimation—and concentrates errors closer to zero. In practice, such bias control and tail suppression are valuable for operational planning, yielding more reliable site-level forecasts even when mean RMSE differences are small.



(a) 20% Source Station – 80% Unseen Station

(b) 40% Source Station – 60% Unseen Station

(c) 60% Source Station – 40% Unseen Station

(d) 80% Source Station – 20% Unseen Station

Figure 7 Residual distribution comparison between VisionEV and LightGBM

Together, the full-shot, few-shot, and zero-shot evaluations show that VisionEV not only performs strongly under full supervision but also adapts with sparse labels and generalizes to novel environments. While margins over a strong tabular baseline (LightGBM) are modest on full-shot splits, VisionEV delivers more reliable accuracy under spatial domain shift and limited-label regimes—the conditions that mirror deployment to new stations or areas. Moreover, compared with LightGBM, VisionEV reduces systematic bias—mitigating both over- and under-estimation—and concentrates errors more tightly around zero (as evidenced by the residual distributions). In planning workflows, this robustness, together with constraint-consistent non-negative integer outputs (without ad-hoc clipping) and imagery-driven representations that obviate handcrafted spatial features, can change top-K rankings and threshold decisions even when aggregate RMSE/MAE differences are small.

4.4 Ablation study

To evaluate the individual contributions of each core component in VisionEV, we conduct ablation experiments by systematically removing one element at a time. Specifically, we assess four model variants: (1) w/o Satellite Image: eliminate the visual modality and use structured metadata only; (2) w/o Prompting: remove the natural language prompt while keeping all feature inputs; (3) w/o Fine-tuning: use the original pretrained PaliGemma directly; visual encoder and text decoder are frozen and the model performs zero-shot inference using the station image plus the structured-to-text prompt, outputting a single numeric token; and (4) w/o Pre-training: initialize the model with random weights and train from scratch. All ablation tests results are reported across full-shot, few-shot, and zero-shot scenarios. The few-shot metrics are averaged over 5%, 10%, 15%, and 20% of the training data, while zero-shot metrics are averaged over four station sampling ratios: 20%, 40%, 60%, and 80%. The detailed results are summarized in Table 4 .

From the table, we observe that removing any one component from VisionEV consistently leads to a decline in predictive performance. The removal of pre-training causes the most severe degradation across all scenarios, with RMSE and MAE increasing to 9.57 and 7.02, respectively, on average. This confirms the foundational role of LLM pre-training in equipping the model with generalized language and reasoning capabilities, especially critical in low-resource (few-shot/zero-shot) settings. Similarly, disabling fine-tuning results in a significant performance drop (average RMSE: 4.46, MAE: 3.34), demonstrating the necessity of downstream adaptation for aligning the model with domain-specific EV charging behavior.

Removing the prompting mechanism also reduces performance (RMSE: 3.43, MAE: 2.43), though less drastically. This highlights the importance of natural language prompts in helping the model structure input features such as traffic volume, and seasonal factor in a way that aligns with the LLM's pretraining distribution. This ablation also serves as a quantitative robustness check for prompt sensitivity, demonstrating that VisionEV's performance is materially influenced by the presence and formulation of textual prompts. Finally, excluding the satellite image input leads to moderate declines (RMSE: 3.95, MAE: 2.84), confirming that visual spatial information provides valuable complementary signals—such as land use, surrounding infrastructure, and urban density—that are difficult to encode with tabular features alone. This result not only highlights the complementary value of spatial visual information but also serves as a quantitative verification of the relevance of unstructured visual features extracted by the SigLIP encoder, confirming that the imagery input meaningfully contributes to VisionEV's predictive performance.

Overall, the ablation study validates the synergistic integration of each component in VisionEV. Satellite imagery provides perceptual spatial grounding, prompt design introduces semantic structure, fine-tuning enables domain adaptation, and pre-training serves as the foundational learning prior. The full model, combining all four, achieves the best performance across all evaluation settings, substantiating the design choices of our multimodal framework.

Table 4 Ablation results

Model Variant	**RMSE**				**MAE**			
Metric	Full	Few	Zero	Mean	Full	Few	Zero	Mean
w/o Satellite Image	3.29	3.87	4.68	3.95	2.41	2.86	3.26	2.84
w/o Prompting	3.11	3.42	3.76	3.43	2.25	2.39	2.65	2.43
w/o Fine-tuning	4.46	4.46	4.46	4.46	3.34	3.34	3.34	3.34
w/o Pre-training	6.78	10.83	11.11	9.57	5.04	7.88	8.15	7.02
VisionEV	**2.87**	**3.30**	**3.48**	**3.22**	**1.98**	**2.29**	**2.59**	**2.29**

4.5 Cross-Region Spatial Hold-Out Validation

To further assess the spatial generalization capability of VisionEV, we conducted an additional cross-region spatial hold-out validation within Kansas City. Charging stations were divided into two geographically distinct regions: (1) the downtown core, characterized by dense commercial and mixed-use development, and (2) the suburban periphery, consisting mainly of residential and industrial land uses. Of the 189 public EV charging stations in KCMO, 91 are located in the downtown core (as delineated in Figure 4-b), and the remaining stations are in the suburban periphery. Correspondingly, out of the total 28,566 station-week samples, 13,649 originate from downtown stations and 14,917 from suburban stations.

We trained on one region (e.g., downtown) and evaluated on the other (e.g., suburban), then reversed the split to form a two-fold cross-validation. This design enforces zero spatial overlap between training and test zones, reducing information leakage from nearby stations and inducing a meaningful distribution shift in built form and land-use context—thereby approximating a cross-city deployment within the same metropolitan area.

As summarized in Table 5, VisionEV maintains strong predictive performance under this strict out-of-area evaluation. When trained on downtown stations and tested on suburban areas, VisionEV achieved RMSE = 3.44 and MAE = 2.42, compared with RMSE = 3.52 / MAE = 2.63 for the strongest baseline (LightGBM). When the direction was reversed—training on suburban and testing on downtown—VisionEV obtained RMSE = 3.41 / MAE = 2.40, again outperforming LightGBM (RMSE = 3.46 / MAE = 2.54) as well as all other baselines.

Table 5 Performance under cross-region spatial hold-out validation

Training → Testing Region	Model	RMSE	MAE	Δ RMSE (%) vs Full-shot	Δ MAE (%) vs Full-shot
Downtown → Suburban	Linear Regression	6.83	5.46	+39.1	+51.7
	LightGBM	3.52	2.63	+21.4	+25.8
	GCN	5.73	4.78	+52.0	+71.3
	Gemma (text-only LLM)	3.87	3.15	+21.7	+26.0
	VisionEV	**3.44**	**2.42**	**+19.9**	**+22.2**
Suburban → Downtown	Linear Regression	6.76	5.37	+37.7	+49.2
	LightGBM	3.46	2.54	+19.3	+21.5

	GCN	5.65	4.59	+49.9	+64.5
	Gemma (text-only LLM)	3.81	3.05	+19.8	+22.0
	VisionEV	**3.41**	**2.40**	**+18.8**	**+21.2**

Across both configurations, VisionEV's error degradation relative to its full-shot setting averaged +19.4% (RMSE) and +21.7% (MAE), which is on par with or lower than LightGBM (+20.4% RMSE, +23.7% MAE) and substantially below the degradations observed for Linear Regression and GCN (RMSE/MAE typically >+38% / >+49%). These results indicate that VisionEV is robust to spatial domain shift and can generalize to unseen urban areas. Although both regions lie within the same metropolitan area, the downtown core and suburban periphery differ sharply in built form, land use, and traffic context; training and testing across them therefore approximates a cross-city deployment. Together with the few-shot and zero-shot experiments, this spatial hold-out provides empirical evidence of VisionEV's cross-region transferability and supports its applicability to large-scale, data-sparse infrastructure planning.

4.6 Computational Cost and Deployment Feasibility.

Our fine-tuning remains lightweight: the vision encoder is frozen and only the projector and decoder are updated, enabling short training cycles and modest memory needs. At inference, VisionEV executes a single batched decode that emits one numeric token, so latency is dominated by the visual forward pass and batch size. On a Colab L1 GPU, processing the full validation split (2,856 examples) completed in 413 seconds ($\approx$ 6.92 examples/s, $\approx$ 0.145 s/example). Extrapolating linearly, 10,000 sites would require roughly 24 minutes prior to caching. In practice, we precompute and cache satellite-image embeddings per station (one-time cost), after which weekly/monthly scoring involves only textual refresh (season/week) plus a short decode step, further reducing runtime.

From a hardware perspective, inference is satisfied by $\geqslant$8 GB VRAM, and lightweight fine-tuning is feasible with ~24 GB VRAM. Two deployment patterns align with planning workflows:

1. **Offline batch scoring** for siting/right-sizing, amortizing compute across planning cycles and regions; cached embeddings substantially cut recurring cost.
2. **A lightweight cloud API** (e.g., a small GPU instance) exposing a REST endpoint for on-demand scoring; this integrates readily with existing dashboards and data pipelines.

Operationally, this design lowers total cost of ownership (TCO) by avoiding repeated feature engineering, minimizing re-tuning, and supporting few/zero-shot updates when policy inputs or siting criteria evolve. In practice, the decision ROI is high: even one avoided mis-siting (with six-figure build costs) can outweigh incremental compute, even when average RMSE/MAE gains are modest. If the sole objective is minimum aggregate error on a stationary, fully labeled dataset, a well-tuned tabular model (e.g., LightGBM) may suffice; however, VisionEV offers the greatest value under spatial shift, data sparsity, changing assumptions, or high feature-curation costs—conditions common in scalable, equitable EV-infrastructure planning.

5. Conclusion

This study introduces VisionEV, a multimodal large language model framework designed to forecast EV charging demand by integrating structured planning data and satellite imagery. By formulating the task as a multimodal token generation problem, VisionEV jointly processes natural language prompts—encoding

station-level metadata such as traffic flow and seasonality—and high-resolution satellite images that capture the unstructured spatial context surrounding each charging station. The architecture combines a frozen SigLIP vision encoder with a decoder-only language model and applies a lightweight fine-tuning strategy using a cosine learning rate schedule. This design enables the model to efficiently adapt pretrained representations to the EV demand forecasting domain, learning from both textual and visual signals without requiring handcrafted spatial features or large-scale training from scratch.

Empirical evaluations on a real-world dataset from Kansas City, Missouri, confirm the robustness, adaptability, and accuracy of VisionEV across diverse operational settings. The model consistently outperforms classical statistical methods, advanced deep learning models, and unimodal LLM baselines across full-shot, few-shot, and zero-shot scenarios. In particular, VisionEV demonstrates strong sample efficiency in low-data regimes and reliable generalization to unseen spatial contexts—key requirements for scalable, real-world deployment. Ablation studies further validate the essential contributions of prompt design, satellite imagery, pretraining, and fine-tuning.

While the present case study centers on Kansas City due to the unique availability of synchronized charging logs, traffic indicators, and high-resolution satellite imagery, the VisionEV framework is designed to be globally applicable. Its input modalities and multimodal backbone are broadly obtainable and transferable, and the few-shot, zero-shot, and spatial hold-out validations collectively demonstrate VisionEV's ability to generalize to unseen urban areas. Importantly, this work focuses on quantifying the marginal value of multimodality rather than exhaustively ranking multimodal backbones: we include a text-only LLM baseline (Gemma) using the same prompt (without imagery) and reinforce non-vision baselines (LR/LightGBM/GCN) with station-level land-use indicators, enabling a clearer view of the contribution from imagery beyond structured planning variables. As a next step, we will broaden the benchmark with ViT-initialized and CLIP-based encoders, as well as recent vision-language models, to provide a more comprehensive assessment of performance and robustness across diverse multimodal designs. Future work will also extend the analysis to multiple cities to examine cross-city scalability and domain adaptation explicitly. Taken together, these results position VisionEV as a practical, generalizable, and data-efficient tool for responsive, cost-effective EV-infrastructure planning at scale.

Appendix A – Short-Run Fine-Tuning Strategy

A-1 Learning Rate Scheduling

To perform effective domain adaptation while mitigating overfitting and catastrophic forgetting, we adopt a short-run fine-tuning approach coupled with a cosine learning rate scheduler. Cosine learning rate decay is chosen for its proven effectiveness in stabilizing training of large transformer-based models. Specifically, the learning rate schedule gradually decreases from an initial peak after a brief warm-up phase, mathematically described as: Let η denote the initial base learning rate, E the total number of epochs, and α the proportion of total training steps dedicated to learning rate warm-up. The learning rate at training step t is computed as the equation below:

$$\eta_t = \begin{cases} \eta * \dfrac{t}{\alpha T}, t \leq \alpha T \ (\text{Warm} - \text{up phase}) \\ \dfrac{\eta}{2}\left[1 + \cos\left(\dfrac{\pi(t - \alpha T)}{(1-\alpha)T}\right)\right], t > \alpha T \ (\text{Cosine decay phase}) \end{cases} \quad \text{Eq. (12)}$$

where $T = \lceil |\, Dtrain \,| / B \rceil \times E$ is the total number of training steps calculated from the training dataset D^{train}, batch size B, and the number of epochs E.

A-2 Parameter Update via SGD

At each training step, batches of multimodal data samples (B_t) are fed through the model, predictions are generated, and gradients of the Negative Log-Likelihood (NLL) loss $\mathcal{L}(\theta)$ (defined in Eq. (11)) are computed with respect to the tunable parameters θ as below):

$$\nabla_\theta \mathcal{L}(\theta) = \frac{\partial}{\partial \theta}\left(-\frac{1}{|B_t|} \sum_{(i,w)\in B_t} logP(y_{i,w}|\mathbf{Z}_{i,w};\theta) \right) \quad \text{Eq. (13)}$$

These gradients are applied via stochastic gradient descent (SGD) using the dynamically adjusted learning rate η_t:

$$\theta_t = \theta_{t-1} - \eta_t \nabla_\theta \mathcal{L}(\theta_{t-1}) \quad \text{Eq. (14)}$$

A-3 Periodic Validation and Model Monitoring

Throughout fine-tuning, model performance is periodically evaluated on a held-out validation dataset D^{val}. This enables continuous monitoring of generalization capability and ensures robust predictions under unseen spatial contexts. Validation assessments occur every Validation Frequency (F_{val}), providing insight into model convergence and preventing overfitting.

A-4 Pseudocode

Short-Run Fine-Tuning with Cosine Learning Rate Scheduling

Require:
Pre-trained model with parameters θ_0 ; training dataset D^{train}; validation dataset D^{val}; batch size B; base learning rate η; warm-up percentage α; number of training epochs E

1. Initialize global training step $t = 0$ and compute total training steps $T = \lceil |\ Dtrain\ |/B \rceil \times E$.
2. Freeze vision encoder parameters $\phi_v(\cdot)$, keep tunable: $\mathbf{W}_v$, $\phi_t(\cdot)$, θ
3. Initialize cosine learning rate schedule:

$$\eta_t = CosineDecayScheduler(\eta, T, \alpha)$$

4. Initialize training data iterator: $I \leftarrow Iterator\left(D^{train}\right)$
5. **for** each epoch $e = 1$ to E **do**
    for each batch $B_t \sim I$ **do**
        Increment global step: $t \leftarrow t + 1$
        Compute current learning rate η_t
        Compute loss and gradients
    Update tunable model parameter via SGD
    Log training loss and current learning rate.
    if $t\ mod\ F_{val} = 0$ validate model:
        Sample a validation batch $B^{val} \sim D^{val}$
        Evaluate validation predictions and log results
    end for
end for

Return: Final fine-tuned model parameters θ_T

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